

MICHAEL G. FAMULARE

Curriculum Vitae

Institute for Disease Modeling
Global Health | Gates Foundation
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Employment

Institute for Disease Modeling Global Health Gates Foundation	Seattle, WA
Principal Research Scientist	Dec 2023 – Present
Senior Research Manager	Jul 2020 – Nov 2023

Institute for Disease Modeling Global Good Intellectual Ventures	Bellevue, WA
Principal Research Scientist, co-chair of Epidemiology	Jan 2019 – Jul 2020
Senior Research Scientist	Jan 2016 – Dec 2018
Research Scientist	May 2012 – Dec 2015

Education

University of Washington	Seattle, WA
Ph.D. in Physics	May 2012
Thesis title: Probabilistic Neural Coding from Deterministic Neural Dynamics—mathematics and biophysics of adaptive single neuron computation.	
New York University	New York, NY
B.S. in Physics Education	May 2004

Professional Service

COVID-19 advisor to Washington State Department of Health and Gov. Inslee	
DOH Modeling Group & Weekly DOH-IDM Modeling Briefing	Apr 2020 – Jul 2021

SCAN: the greater Seattle Coronavirus Assessment Network	
co-PI at-home COVID-19 community testing program	Mar 2020 – Jul 2021

Global Polio Eradication Initiative	
nOPV2 Working Group: clinical/pilot studies sub-group	Jan 2020 – Mar 2020
Post-Cessation Polio Outbreak Strategic Simulation Exercise	Sep 2018
Cessation Risk Task Team (CRTT)	Jul 2016 – Mar 2020

IDM Symposium	
Session organizer: Evolutionary Epidemiology	Apr 2018
Session organizer: Surveillance	Apr 2016

SIAM LS10 Minisymposium	
Understanding the Link Between Neuronal Dynamics and Computation	Jul 2010

University of Washington Department of Physics	
12X Labs Subcommittee	Jun 2009 - Jun 2010

Professional Development

Harvard Extension School	Cambridge, MA
Managing Yourself and Leading Others	June 2019
Leading More Effective Teams	June 2019
Summer Institute in Statistics and Modeling in Infectious Diseases	Seattle, WA
Evaluating Immune Correlates of Protection	July 2014
Evolutionary Dynamics and Molecular Epidemiology of Viruses	July 2012
Stochastic Epidemic Models with Inference	July 2012
MCMC1 for Infectious Diseases	July 2012
Contact Network Epidemiology	July 2012
Kavli Institute for Theoretical Physics	Santa Barbara, CA
Emerging Techniques in Neuroscience	Sep 2010 – Nov 2010
Marine Biological Laboratory	Woods Hole, MA
Methods in Computational Neuroscience	Aug 2009

Preprints

For most recent, see my medRxiv and biorXiv author searches.

- [7] Núñez RC, Hart GR, **Famulare M**, Lorton C, Herbeck JT (2024). Using phylogenetic summary statistics for epidemiological inference. *bioRxiv* 2024.08.07.607080. doi:10.1101/2024.08.07.607080; [local PDF]
- [6] Thakkar N and **Famulare M** (2023). A generating function perspective on the transmission forest. *arXiv* 2311.16317. doi:10.48550/arXiv.2311.16317; [local PDF]
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- [4] Burstein R, Althouse BM, Adler A, Akullian A, Brandstetter E, Cho S, Emanuels A, Fay K, Gamboa L, Han P, Huden K, Ilcisin M, Izzo M, Jackson ML, Kim AE, Kimball L, Lacombe K, Lee J, Logue JK, Rogers J, Chung E, Sibley TR, Van Raay K, Wenger E, Wolf CR, Boeckh M, Chu H, Duchin J, Rieder M, Shendure J, Starita LM, Viboud C, Bedford T, Englund JA, **Famulare M** (2022). Interactions among 17 respiratory pathogens: a cross-sectional study using clinical and community surveillance data. *medRxiv* 2022.02.04.22270474. doi:10.1101/2022.02.04.22270474; [local PDF]
- [3] Cohen JA, Stuart RM, Rosenfeld K, Lyons H, White M, Kerr CC, Klein DJ, **Famulare M** (2021). Quantifying the role of naturally- and vaccine-derived neutralizing antibodies as a correlate of protection against COVID-19 variants. *medRxiv* 2021.05.31.21258018. doi:10.1101/2021.05.31.21258018; [local PDF]; [supplement 1]
- [2] Chao DL, Oron AP, Srikrishna D, **Famulare M** (2020). Modeling layered non-pharmaceutical interventions against SARS-CoV-2 in the United States with Corvid. *medRxiv* 2020.04.08.20058487. doi:10.1101/2020.04.08.20058487; [local PDF]
- [1] **Famulare M**, Fairhall A (2011). Adaptive probabilistic neural coding from deterministic spiking neurons: analysis from first principles. *arXiv* doi:10.48550/arXiv.1111.0097; [local PDF]

Journal Articles

For most recent, see my Google Scholar page.

- [34] Perofsky AC, Hansen CL, Burstein R, Boyle S, Prentice R, Marshall C, Reinhart D, Capodanno B, Truong M, Schwabe-Fry K, Kuchta K, Pfau B, Acker Z, Lee J, Sibley TR, McDermot E, Rodriguez-Salas L, Stone J, Gamboa L, Han PD, Adler A, Waghmare A, Jackson ML, **Famulare M**, Shendure J, Bedford T, Chu HY, Englund JA, Starita LM, Viboud C (2024). Impacts of human mobility on the citywide transmission dynamics of 18 respiratory viruses in pre- and post-COVID-19 pandemic years. *Nature Communications* 15:4164. doi:10.1038/s41467-024-48528-2; [local PDF]
- [33] Streicker DG, Griffiths ME, Antia R, Bergner L, Bowman P, dos Santos de Moraes MV, Esvelt K, **Famulare M**, Gilbert A, He B, Jarvis MA, Kennedy DA, Kuzma J, Nasimiyyu Wanyonyi C, Remien C, Rocke T, Rosenke K, Schreiner C, Sheen J, Simons D, Yordanova IA, Bull JJ, and Nuismer SL (2024). Developing transmissible vaccines for animal infections: Intrinsically safe designs and a staged transparent development process will be essential. *Science* 384:6693. doi:10.1126/science.adn3231; [local PDF]
- [32] Wong W, Gauld J, **Famulare M** (2023). From Vaccine to Pathogen: Modeling Sabin 2 Vaccine Virus Reversion and Evolutionary Epidemiology. *Virus Evolution* 9(2):vead044. doi:10.1093/ve/vead044; [local PDF]; [supplement 1]
- [31] Cohen JA, Stuart RM, Panovska-Griffiths J, Mudimu E, Abeyasuriya RG, Kerr CC, **Famulare M**, Klein DJ (2023). The changing impact of vaccines in the COVID-19 pandemic. *Cell Reports* S2211-1247(23)00319-4. doi:10.1016/j.celrep.2023.112308; [local PDF]; [supplement 1]
- [30] Hansen C, Perofsky AC, Burstein R, **Famulare M**, Boyle S, Prentice R, Marshall C, McCormick BJJ, Reinhart D, Capodanno B, Truong M, Schwabe-Fry K, Kuchta K, Pfau B, Acker Z, Lee J, Sibley TR, McDermot E, Rodriguez-Salas L, Stone J, Gamboa L, Han PD, Duchin JS, Waghmare A, Englund JA, Shendure J, Bedford T, Chu HY, Starita LM, Viboud C (2022). Trends in Risk Factors and Symptoms Associated With SARS-CoV-2 and Rhinovirus Test Positivity in King County, Washington, June 2020 to July 2022. *JAMA Network OPEN* 5(12):e2245861. doi:10.1001/jamanetworkopen.2022.45861; [local PDF]
- [29] Brouwer AF, Eisenberg MC, Shulman LM, **Famulare M**, Koopman JS, Kroiss SJ, Hindiyyeh M, Manor Y, Grotto I, Eisenberg JNS (2022). The role of time-varying viral shedding in modelling environmental surveillance for public health: revisiting the 2013 poliovirus outbreak in Israel. *J Roy Soc Interface* 19:20220006. doi:10.1098/rsif.2022.0006; [local PDF]
- [28] Paredes MI, Lunn SM, **Famulare M**, Frisbie LA, Painter I, Burstein R, Roychoudhury P, Xie H, Bakhash SA, Perez R, Lukes M, Ellis S, Sathees S, Mathias PC, Greninger A, Starita LM, Frazar CD, Ryke E, Zhong W, Gamboa L, Threlkeld M, Lee J, Nickerson DA, Bates DL, Hartman ME, Haugen E, Nguyen TN, Richards JD, Rodriguez JL, Stamatoyannopoulos JA, Thorland E, Melly G, Dykema PE, MacKellar DC, Gray HK, Singh A, Peterson JM, Russel D, Torres LM, Lindquist S, Bedford T, Allen KJ, Oltean H (2021). Associations between SARS-CoV-2 variants and risk of COVID-19 hospitalization among confirmed cases in Washington State: a retrospective cohort study. *Clinical Infectious Diseases* ciac279. doi:10.1093/cid/ciac279; [local PDF]; [supplement 1]
- [27] **Famulare M**, Wong W, Haque R, Platts-Mills JA, Saha P, Aziz AB, Ahmed T, Islam MO, Uddin MJ, Bandyopadhyay AS, Yunus M, Zaman Q, Taniuchi M (2021). Multiscale model for forecasting Sabin 2 vaccine virus household and community transmission. *PLOS Computational Biology* 17(2):e1009690. doi:10.1371/journal.pcbi.1009690; [local PDF]
- [26] Jackson ML, Hart GR, McCulloch DJ, Adler A, Brandstetter E, Fay K, Han P, Lacombe K, Lee J, Sibley TR, Nickerson DA, Rieder MJ, Starita L, Englund JA, Bedford T, Chu H, **Famulare M** (2021). Effects of weather-related social distancing on city-scale transmission of respiratory viruses: a retrospective cohort study. *BMC Infectious Diseases* 21(1):335. doi:10.1186/s12879-021-06028-4; [local PDF]
- [25] Kerr CC, Stuart RM, Mistry D, Abeyasuriya R, Rosenfeld K, Hart GR, Nuñez RC, Cohen JA, Selvaraj P, Hagedorn B, George L, Jastrzebski M, Izzo AS, Fowler G, Wagner A, Chang

- ST, Oron AP, Wenger EA, Panovska-Griffiths J, **Famulare M**, Klein DJ (2021). Covasim: an agent-based model of COVID-19 dynamics and interventions. *PLOS Computational Biology* 17(7):e1009149. doi:10.1371/journal.pcbi.1009149; [local PDF]
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- [23] Tordoff DM, Greninger AL, Roychoudhury P, Shrestha L, Xie H, Jerome KR, Breit N, Huang ML, **Famulare M**, Herbeck JT (2021). Phylogenetic estimates of SARS-CoV-2 introductions into Washington State. *Lancet Regional Health* 1:100018. doi:10.1016/j.lana.2021.100018; [local PDF]; [supplement 1]
- [22] Goyal A, Reeves DB, Thakkar N, **Famulare M**, Cardozo-Ojeda EF, Mayer BT, Schiffer JT (2021). Slight reduction in SARS-CoV-2 exposure viral load due to masking results in a significant reduction in transmission with widespread implementation. *Scientific Reports* 11(1):11838. doi:10.1038/s41598-021-91338-5; [local PDF]
- [21] Müller NF, Wagner C, Frazar CD, Roychoudhury P, Lee J, Moncla LH, Pelle B, Richardson M, Ryke E, Xie H, Shrestha L, Addetia A, Rachleff VM, Lieberman NAP, Huang ML, Gautom R, Melly G, Hiatt B, Dykema P, Adler A, Brandstetter E, Han PD, Fay K, Ilcisin M, Lacombe K, Sibley TR, Truong M, Wolf CR, Boeck M, Englund JA, **Famulare M**, Lutz BR, Rieder MJ, Thompson M, Duchin JS, Starita LM, Chu HY, Shendure J, Jerome KR, Lingquist S, Greninger AL, Nickerson DA, Bedford T (2021). Viral genomes reveal patterns of the SARS-CoV-2 outbreak in Washington State. *Science Translational Medicine* 13(595):eabf0202. doi:10.1126/scitranslmed.abf0202; [local PDF]
- [20] Kerr CC, Mistry D, Stuart RM, Rosenfeld K, Hart GR, Nuñez RC, Cohen JA, Selvaraj P, Abeyesuriya RG, Jastrzebski M, George L, Hagedorn B, Panovska-Griffiths J, Fagalde M, Duchin J, **Famulare M**, Klein DJ (2021). Controlling COVID-19 via test-trace-quarantine. *Nature Communications* 12(1):2993. doi:10.1038/s41467-021-23276-9; [local PDF]
- [19] Jackson ML, Starita L, Biniry E, Phillips CH, Wellwood S, Cho S, Kiavand A, Truong M, Han P, Richardson M, Wolf CR, Heimonen J, Nickerson DA, Chu HY, Boeckh M, Englund JA, **Famulare M**, Lutz BR, Rieder MJ, Thompson M, Shendure J, Bedford T, Adler A, Brandstetter E, Bosua J, Frazar CD, Han PD, Gulati RK, Hadfield J, Huang S, Kimball LE, Lacombe K, Logue JK, Lyon V, Newman KL, Sibley TR, Zigman Suchsland ML, Seattle Flu Study Investigators (2021). Incidence of medically attended acute respiratory illnesses due to respiratory viruses across the life course during the 2018/19 influenza season. *Clinical Infectious Diseases* 73(5). doi:10.1093/cid/ciab131; [local PDF]
- [18] Valesano AL, Taniuchi M, Fitzsimmons WJ, Islam MO, Ahmed T, Zaman K, Haque R, Wong W, **Famulare M**, Lauring AS (2021). The early evolution of oral poliovirus vaccine is shaped by strong positive selection and tight transmission bottlenecks. *Cell Host & Microbe* 29(1). doi:10.1016/j.chom.2020.10.011; [local PDF]
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- T, Greninger A, Roychoudhury P, **Famulare M**, Chu HY, Shendure J, Starita L, Anderson C, Gangavarapu K, Zeller M, Spencer E, Andersen K, MacCannell D, Tong S, Armstrong G, Paden C, Li Y, Zhang Y, Morrow S, Willis M, Matyas B, Mase S, Kasirye O, Park M, Chan C, Yu A, Chai S, Villarino E, Bonin B, Wadford D, and Chiu CY (2020). A Genomic Survey of SARS-CoV-2 Reveals Multiple Introductions into Northern California without a Predominant Lineage. *Science* 369(6503). doi:10.1126/science.abb9263; [local PDF]
- [15] CDC COVID Response Team, Jorden MA, Rudman SL, Villarino E, Hoferka S, Patel MT, Bemis K, Simmons CR, Jespersen M, Johnson JI, Mytty E, Arends KD, Henderson JJ, Mathes RW, Weng CX, Duchin J, Lenahan J, Close N, Bedford T, Boeckh M, Chu HY, Englund JA, **Famulare M**, Nickerson DA, Rieder MJ, Shendure J, Starita LM, Armstrong GL, Butler JC, Coletta MA, Kite-Powell A, Bhatnagar J, Reagan-Steiner S, Tong S, Flannery B, Ferdinands JM, Chung JR (2020). Evidence for limited early spread of COVID-19 within the United States, January–February 2020. *MMWR* 69(22). doi:10.15585/mmwr.mm6922e1; [local PDF]
- [14] Chu HY, Englund JA, Starita LM, **Famulare M**, Brandstetter E, Nickerson DA, Rieder MJ, Adler A, Lacombe K, Kim AE, Graham C, Logue J, Wolf CR, Heimonen J, McCulloch DJ, Han PD, Sibley TR, Lee J, Ilcisin M, Fay K, Burstein R, Martin B, Lockwood CM, Thompson M, Lutz B, Jackson M, Hughes JP, Boeckh M, Shendure J, Bedford T (2020). Early detection of Covid-19 through a citywide pandemic surveillance platform. *New England Journal of Medicine* 383(2). doi:10.1056/NEJMc2008646; [local PDF]
- [13] Chu HY, Boeckh M, Englund JA, **Famulare M**, Lutz BR, Nickerson DA, Rieder MJ, Starita LM, Adler A, Brandstetter E, Frazar CD, Han PD, Gulati RK, Hadfield J, Jackson ML, Kiavand A, Kimball LE, Lacombe K, Logue J, Lyon V, Newman KL, Sibley TR, Zigman Suschland ML, Wolf C, Shendure J, and Bedford T (2020). Protocol: The Seattle Flu Study: a multi-arm community-based prospective study protocol for assessing influenza prevalence, transmission, and genomic epidemiology. *BMJ Open* 10(10):e037295. doi:10.1136/bmjopen-2020-037295; [local PDF]
- [12] Kroiss SJ, Ahmadzai M, Ahmed J, Alam MM, Chabot-Couture G, **Famulare M**, Mahamud A, McCarthy KA, Mercer LD, Muhammad S, Safdar RM, Sharif S, Shaukat S, Shukla H, Lyons H (2018). Assessing the sensitivity of the polio environmental surveillance system. *PLOS One* 13(12):e0208336. doi:10.1371/journal.pone.0208336; [local PDF]
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- [10] McCarthy KA, Chabot-Couture G, **Famulare M**, Lyons HM, Mercer LD (2017). The risk of type 2 oral polio vaccine use in post-cessation outbreak response. *BMC Medicine* 15(1):175. doi:10.1186/s12916-017-0937-y; [local PDF]
- [9] Kroiss SJ, **Famulare M**, Lyons H, McCarthy KA, Mercer LD, Chabot-Couture G (2017). Evaluating cessation of the type 2 oral polio vaccine by modeling pre-and post-cessation detection rates. *Vaccine* 35(42):5674–5681. doi:10.1016/j.vaccine.2017.08.048; [local PDF]; [supplement 1]
- [8] Taniuchi M, **Famulare M**, Zaman K, Uddin MJ, Upfill-Brown AM, Ahmed T, Saha P, Haque R, Bandyopadhyay AS, Modlin JF, Platts-Mills JA, Houpt ER, Yunus M, and Petri WA (2017). Community transmission of type 2 poliovirus after cessation of trivalent oral polio vaccine in Bangladesh: an open-label cluster-randomised trial and modelling study. *Lancet Infectious Diseases* 17(1):1069–1079. doi:10.1016/S1473-3099(17)30358-4; [local PDF]
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- [5] **Famulare M** and Hu H (2015). Extracting transmission networks from phylogeographic data for epidemic and endemic diseases: Ebola virus in Sierra Leone, 2009 H1N1 pandemic influenza and polio in Nigeria. *International Health* 7(2):130–138. doi:10.1093/inthealth/ihv031; [local PDF]
- [4] Mease RA, **Famulare M**, Gjorgjieva J, Moody WJ, and Fairhall AL (2013). Emergence of adaptive computation by single neurons in the developing cortex. *Journal of Neuroscience* 33(30):12154–12170. doi:10.1523/JNEUROSCI.3263-12.2013; [local PDF]
- [3] Goldwyn JH, Imennov NS, **Famulare M**, and Shea-Brown E (2011). On stochastic differential equation models for ion channel noise in Hodgkin-Huxley neurons. *Physical Review E* 83:041908. doi:10.1103/PhysRevE.83.041908; [local PDF]
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Public Health Reports

- [53] Singh G, Painter I, **Famulare M**, Thakkar N, Lavista Ferres JM, Etzioni R, Richardson B, Wasserman C (21 Jul 2021). WA Situation Report 37: COVID-19 transmission across Washington State. (doh.wa.gov); [local PDF]
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- [50] Singh G, Painter I, **Famulare M**, Thakkar N, Lavista Ferres JM, Etzioni R, Richardson B, Wasserman C (2 Jun 2021). WA Situation Report 34: COVID-19 transmission across Washington State. (doh.wa.gov); [local PDF]
- [49] Singh G, Painter I, **Famulare M**, Thakkar N, Lavista Ferres JM, Etzioni R, Richardson B, Wasserman C (4 May 2021). WA Situation Report 33: COVID-19 transmission across Washington State. (doh.wa.gov); [local PDF]
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- [47] Singh G, Painter I, **Famulare M**, Thakkar N, Lavista Ferres JM, Etzioni R, Richardson B, Wasserman C (20 Apr 2021). WA Situation Report 31: COVID-19 transmission across Washington State. (doh.wa.gov); [local PDF]
- [46] Singh G, Painter I, Thakkar N, **Famulare M**, Lavista Ferres JM, Etzioni R, Richardson B, Wasserman C (27 Jan 2021). WA Situation Report 25: COVID-19 transmission across Washington State. (doh.wa.gov); [local PDF]
- [45] Nuñez RC, **Famulare M**, Seaman V, and Klein DJ (27 Jan 2021). In search of an effective NPI blend for COVID-19 control in rural environments: lessons learned from the early epidemic control measures in Yakima County, Washington. (covid.idmod.org); [local PDF]

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Conference Proceedings

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Selected Media

BBC Two. 54 days: America and the Pandemic	2 Feb 2021
King 5 News. Don't bring COVID-19 to dinner: Washington health officials warn against holiday gatherings	18 Nov 2020
This Podcast Will Kill You. COVID-19 Chapter 11: Modeling (transcript)	4 May 2020
The New Yorker. Seattle's Leaders Let Scientists Take the Lead. New York's Did Not	4 May 2020
The New Yorker. The End of the Beginning: Seattle Braces for the Next Phase of the Coronavirus Fight	4 May 2020
Public Health Seattle & King County. Transcript of a News Teleconference	14 Apr 2020
The Seattle Times. How big will the coronavirus outbreak get? This Bellevue scientist is figuring that out	15 Mar 2020
The New York Times. Coronavirus May Have Spread in U.S. for Weeks, Gene Sequencing Suggests	1 Mar 2020

Invited Talks

The Foundations of Biomedical Data Science. Choices Give Meaning to Uncertainty: Stories from Pandemic Emergence, Policy Advising, and (very) Personalized Medicine. (recording)	Mar 2023
Metaculus Million Predictions Hackathon. The Best Tools are Fit for Purpose: A Perspective on Information Flow, Modeling, and Decision Making. (recording)	Dec 2022
Washington State Legislature, Democratic Caucus: COVID-19 Briefing-Schools	Mar 2021
Washington State Senate: Health and Long-term Care Committee	Sep 2020
Challenge Seattle COVID-19 Briefing	Aug 2020
Washington State Legislature, Democratic Caucus: COVID-19 Briefing-SitRep	Jul 2020
Challenge Seattle COVID-19 Briefing	Jul 2020
Enabling infectious disease genomics for global health, Cambridge MA	Jun 2018
BC CDC Grand Rounds. Biological Challenges on the Path to Poliovirus Eradication.	May 2018
Vancouver Bioinformatics Users Group (VanBUG). Evolutionary epidemiology of vaccine-derived poliovirus transmission. (recording)	May 2018
Malaria Modeling Consortium, Seattle WA	May 2018
University of Michigan School of Public Health Seminar, Ann Arbor MA	Dec 2017
Langer Lab Seminar (MIT), Cambridge MA	Mar 2017
Panel: Exploring Life Sciences Careers, Boston MA	Feb 2017
PyLadies, Seattle WA	Jul 2016
WHO Polio Research Council, Seattle WA	Apr 2016
IDM Symposium, Bellevue WA	Apr 2016
MAC-EPID Symposium at the University of Michigan, Ann Arbor WA	Mar 2016
Fred Hutchinson Cancer Research Institute VISC Analysis meeting, Seattle WA	Apr 2015
Consortium for a New OPV, Napa CA	Feb 2014
Third Northwest Computational Neuroscience Connection, Seattle WA	Sep 2011
Methods in Computational Neuroscience, Woods Hole MA	Aug 2011
Seminar: Group for Neural Theory at l'École Normale Supérieure, Paris FR	Feb 2011
Emerging Techniques in Neuroscience, Santa Barbara CA	Oct 2010
SIAM Life Sciences, Pittsburgh PA	Jul 2010
Mathematical Biology Seminar, Davis CA	May 2010

Conference Poster Sessions and Talks

- [11] **Famulare M.** Experience with modeling to inform COVID-19 control strategy in Washington State. Second Symposium in Global Health: Shanghai Science and Technology Exchange Center 2020.
- [10] **Famulare M.** High-resolution mapping of respiratory pathogens in the Seattle Metro Area. *Epidemics* 7, 2019.
- [9] **Famulare M.** Beyond reversion: evolutionary epidemiology of vaccine-derived poliovirus transmission. II Joint Congress on Evolutionary Biology 2018.
- [8] **Famulare M,** Chang S, Behrend M, Stern A, Iber J, Zhao K, Adeniji A, Baba M, Burns CC, Obserste MS. Analysis of the evolutionary dynamics of live polio vaccine in cases of non-polio acute flaccid paralysis in Nigeria. NIH Translational and Regulatory Science of Polio Vaccine and Antivirals 2014.
- [7] **Famulare M,** Hu H, and Behrend M. Phylogeography of infectious diseases by direct descent: a pairwise method to reconstruct transmission history from genetic sequence data. *Epidemics*⁴ 2013.
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- [3] **Famulare M**, Mease RA, and Fairhall AL. Origins of contrast gain control in isolated cortical neurons: deriving the code from the dynamics. COSYNE 2010.
- [2] **Famulare M**, Wark B, Mease RA, and Fairhall AL. Mechanisms of adaptation to stimulus statistics in neuronal systems (talk). APS March Meeting 2010.
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Research Support

The Institute for Disease Modeling (IDM) is a research institute within the Global Health division of the Bill and Melinda Gates Foundation. My role is fully funded within an organization that takes no external funding. Prior to being incorporated into the Gates Foundation in July 2020, IDM was an institute within Global Good at Intellectual Ventures. My role there was also fully funded, and I received no external funding. The following represent a selection of projects on which I was/am a co-Investigator and/or a co-author of the funded proposal and for which my leadership was essential to receiving funding.

Environmental surveillance for typhoid in Nepal. Bill & Melinda Gates Foundation. Co-investigator with PI Eric Alm (Massachusetts Institute of Technology). Jan 2019

Seattle Flu Study. Gift from Gates Ventures. Co-Principal Investigator. Nov 2018

Purpose: to produce evidence for the improved community-level safety of genetically-stabilized live-attenuated polio vaccines. Bill & Melinda Gates Foundation. Co-investigator with PI Mami Taniuchi (University of Virginia). Aug 2018

Purpose: to fill in existing knowledge gaps of the dynamics of Sabin virus transmission at a population level and on the impact of different vaccination schedules on such potential risk of transmission. Bill & Melinda Gates Foundation. Co-investigator with PI William Petri (University of Virginia). Dec 2014

Teaching Experience

Instructor: Methods in Computational Neuroscience Aug 2010 & Aug 2011
Marine Biological Laboratory, Woods Hole, MA
Lecturer: Connections between adaptive computation and single neuron dynamics.

Lead Teaching Assistant: Physics 123 Jun 2009 – Jun 2010 Department of Physics, UW
Introductory Physics sequence labs (1XX)

Teaching Assistant Sep 2005 – Jun 2009 Department of Physics, UW
Introductory Physics sequence lectures and labs (1XX), Light and Color for non-majors (214), undergraduate Quantum Mechanics (324), Electricity and Magnetism (331), and graduate Nuclear Theory (560).

Physics Teacher: Stuyvesant High School Sep 2004 – Jun 2005 New York, NY

Special Education Teaching Assistant: Bergen County Special Services Jan 2003 – Aug 2005 Bergen County, NJ

Student Teacher: The Beacon School

Sep 2003 – Jun 2004 New York, NY

Private Tutor

Feb 2000 – Jun 2005 Boston, MA and New York, NY

Date compiled: August 12, 2026.